

Hoan Nguyen

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Research Interests

Aubry–Mather theory; minimal surfaces; min–max theory; geometric flows.

Education

The University of Chicago

Chicago, IL

- Ph.D. in Mathematics, expected 2027. Advisor: André Neves.
- M.S. in Mathematics, 2022.

Hong Kong University of Science and Technology

Hong Kong

- B.Sc. in Mathematics, 2021.
- Senior thesis: *Kähler–Ricci flow and conformal submersion*. Advisor: Frederick Tsz-Ho Fong.

Awards and Honors

- Gold Medal, International Mathematical Olympiad, 2014 and 2015.
- HKUST Academic Excellence Award.

Publications and Preprints

Differential Geometry and Geometric Analysis

- *Minimal foliations, codimension-one stable norms, and a question of Bangert*. Preprint, 2026. arXiv:2608.18428.
- *A min–max gap characterization of minimal foliations on the torus*. Preprint, 2026. arXiv:2605.12428.
- *Kähler–Ricci flow and conformal submersion*. *Mediterranean Journal of Mathematics*, 2023.

Other Publications

- *Residual entropy of confined and interfacial monolayer water ice*. Joint with Lien Le. *Computational and Theoretical Chemistry*, 2026.

Invited Talks

- *A min–max gap characterization of minimal foliations on a torus and applications*. UIUC Geometric Analysis Seminar, Urbana–Champaign, April 21, 2026.
- Upcoming talk, Lehigh Geometry Seminar, Lehigh University, September 21, 2026.
- Upcoming talk, Analysis and Geometric Analysis Seminar, Cornell University, September 28, 2026.

Teaching Experience

The University of Chicago

- Instructor: Elementary Functions and Calculus I, II, III; Calculus I, II; Linear Algebra.
- Teaching Assistant: Accelerated Analysis; Introduction to Dynamical Systems; Functional Analysis.

Technical Skills

- Programming: Python; R (basic).